

REQUIREMENTS ENGINEERING

Software Engineering Seminar

Author: Eng. Carlos Andrés Sierra, M.Sc.
cavirguezs@udistrital.edu.co

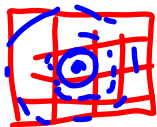
Full-time Adjunct Professor
Computer Engineering Program
School of Engineering
Universidad Distrital Francisco José de Caldas

2025-III

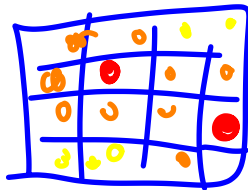


UNIVERSIDAD DISTRITAL
FRANCISCO JOSÉ DE CALDAS

Outline

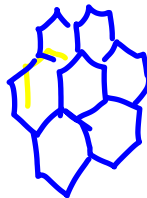


1 Concepts Generation & Selection /



2 Basic Concepts /

3 Requirements Engineering /



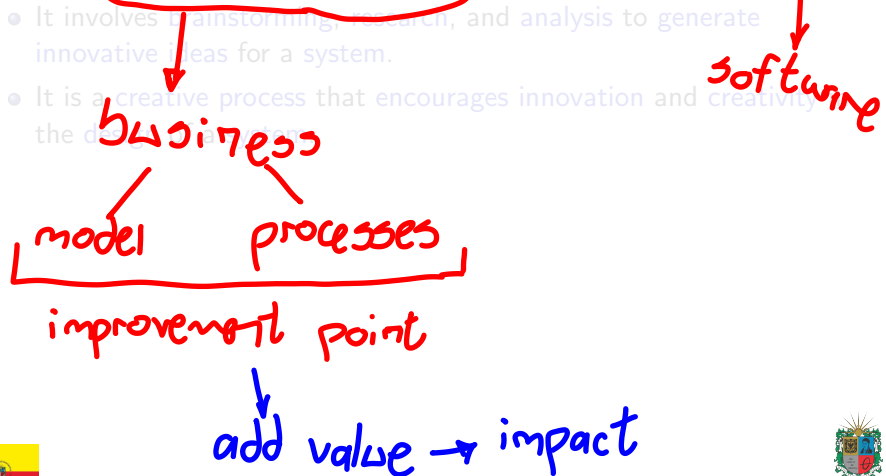
Outline

- 1 Concepts Generation & Selection
- 2 Basic Concepts
- 3 Requirements Engineering



Concepts Generation

- **Concepts generation** is the **process** of **creating ideas** for a **system** that **meet the needs** of its **users**.



Concepts Generation

- **Concepts generation** is the process of creating ideas for a system that meet the needs of its users.
- It involves brainstorming, research, and analysis to generate innovative ideas for a system.
- It is a creative process that encourages innovation and creativity in the design of a system.

patent

netflix



Concepts Generation

- **Concepts generation** is the process of creating ideas for a system that meet the needs of its users.
- It involves brainstorming, research, and analysis to generate innovative ideas for a system.
- It is a creative process that encourages innovation and creativity in the design of a system.

Optimization



Innovation and Creativity

- **Innovation** is the (process) of creating new ideas and solutions that improve the performance of a system.
- Creativity is the ability to generate original and innovative ideas that solve problems and meet the needs of users.
- They are important for ensuring that a system is robust, efficient, and effective.

common

- resources
- quality



Innovation and Creativity

- **Innovation** is the process of creating new ideas and solutions that improve the performance of a system.
- **Creativity** is the ability to generate original and innovative ideas that solve problems and meet the needs of users.
- They are important for ensuring that a system is robust, efficient, and effective.



Innovation and Creativity

- **Innovation** is the process of creating new ideas and solutions that improve the performance of a system.
- **Creativity** is the ability to generate original and innovative ideas that solve problems and meet the needs of users.
- They are important for ensuring that a system is robust, efficient, and effective.

(3)

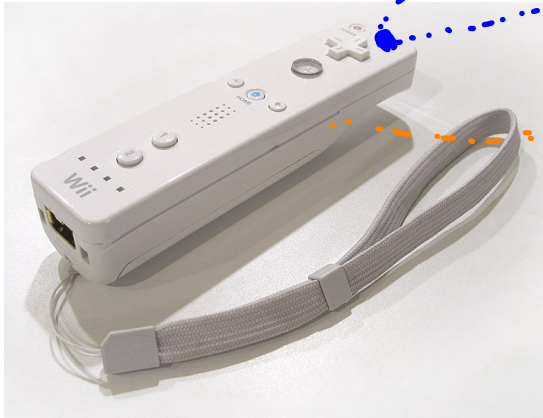
(1)

(2)



Is this Innovation & Creativity?

A real videogames console **revolution**!



Wii

Pitagoras

Bluetooth



Concepts Selection

- **Concepts selection** is the process of evaluating and choosing the best ideas for a system.
- It involves analysis, comparison, and evaluation of concepts to determine which ones are the most feasible and effective.
- It is a critical process that ensures that the final design of a system meets the needs of its users.



Outline

- 1 Concepts Generation & Selection
- 2 Basic Concepts
- 3 Requirements Engineering



Stakeholders Vs. Shareholders

- **Stakeholders** are individuals or groups who have an interest in the success of a project.
- **Stakeholders** can be internal or external to a company. For example, customers, employees, suppliers, and regulators are external stakeholders.
- **Shareholders** are individuals or groups who have an ownership interest in a company.
- **Shareholders** are typically internal to a company. For example, investors, owners, and managers are considered shareholders.



Stakeholders Vs. Shareholders

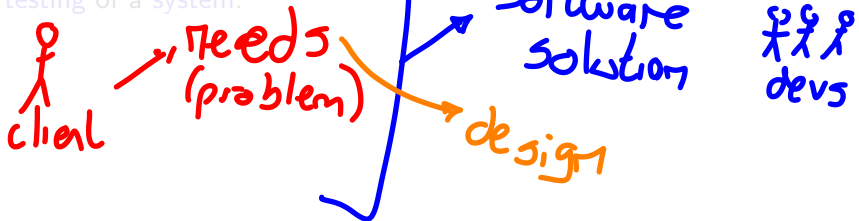
- **Stakeholders** are individuals or groups who have an interest in the success of a project.
- **Stakeholders** can be internal or external to a company. For example, customers, employees, suppliers, and regulators are external stakeholders.
- **Shareholders** are individuals or groups who have an ownership interest in a company.
- **Shareholders** are typically internal to a company. For example, investors, owners, and managers are considered shareholders.

Owners

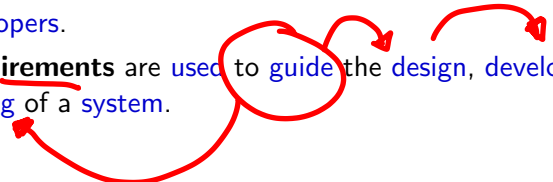


Requirements

- **Requirements** are statements that describe the features, functions, and constraints of a system.
- **Requirements** are used to communicate the needs of stakeholders to developers.
- Requirements are used to guide the design, development, and testing of a system.



Requirements

- **Requirements** are statements that describe the features, functions, and constraints of a system.
 - **Requirements** are used to communicate the needs of stakeholders to developers.
 - **Requirements** are used to guide the design, development, and testing of a system.
- 



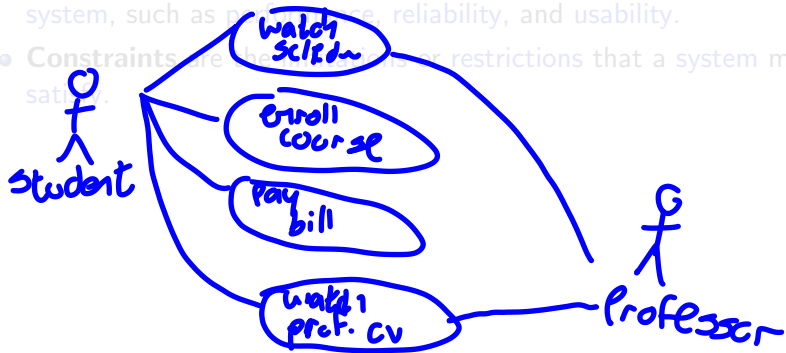
Requirements Types

- **Functional** requirements describe the functions and features of a system.

→ Use cases

- Non-functional requirements describe the quality attributes of a system, such as performance, reliability, and usability.

- Constraints are conditions or restrictions that a system must satisfy.



Requirements Types

- **Functional requirements** describe the **functions** and **features** of a **system**.
- **Non-functional requirements** describe the **quality attributes** of a system, such as **performance**, **reliability**, and **usability**.
- **Constraints** are the limitations or restrictions that a system must satisfy.

scalability



Requirements Types

- **Functional requirements** describe the **functions** and **features** of a **system**.
- **Non-functional requirements** describe the **quality attributes** of a **system**, such as **performance**, **reliability**, and **usability**.
- **Constraints** are the limitations or restrictions that a **system** must satisfy.

business
rules



User Stories

→ close project

- **User stories** are short, simple descriptions of a feature or function of a system.
- They are written from the perspective of the user and describe what the user wants to achieve.
- They are used to capture the requirements of a system in a simple and understandable way.

validate

no technical

Understanding
comprehension
be in the
same page



User Story Format [Example]

User Story

Title: <i>explicit</i>	Priority:	Estimate:
User Story: As a [description of user], I want [functionality] so that [benefit].		
Acceptance Criteria: Given [how things begin] When [action taken] Then [outcome of taking action]		

Planning Poker (orange arrow pointing to Estimate)

desired (pink arrow pointing to User Story)

role (red arrow pointing to [description of user])

req. impact (red text next to [description of user])

test escenario (pink text next to Acceptance Criteria)

expected result (pink text next to [outcome of taking action])

scope (blue curved arrow pointing to the entire User Story box)

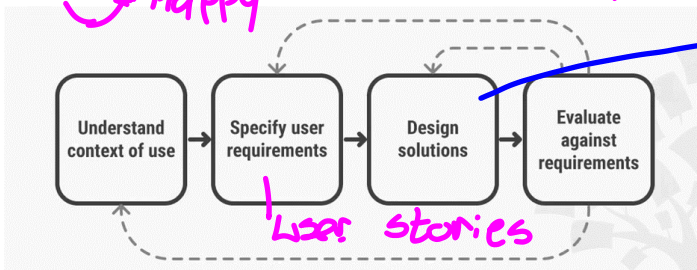
> 1 valid (pink text next to a box containing '1')

ProductPlan

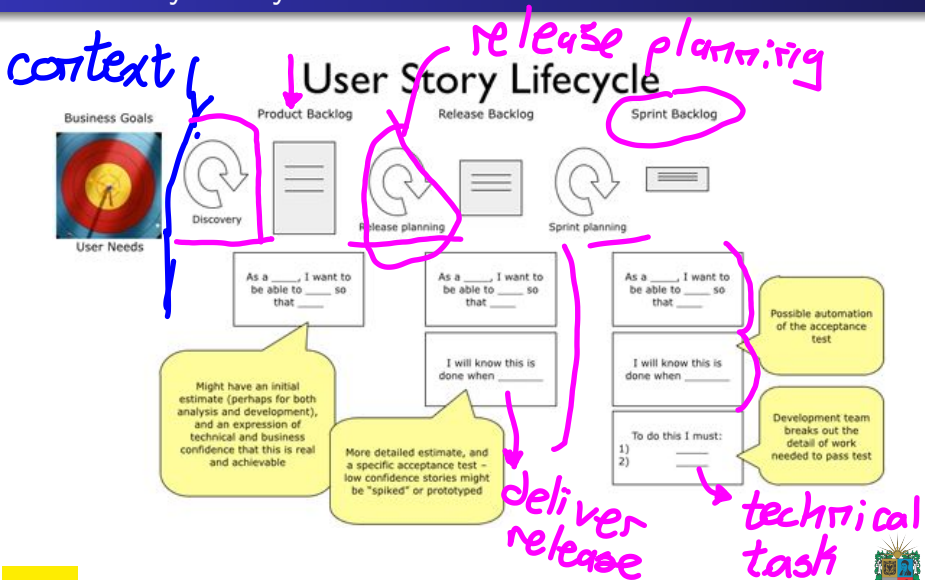


User-Centered Design (UCD)

- **User-centered design** (UCD) is an **iterative** design process that focuses on **understanding** the **needs**, **preferences**, and **behaviors** of users.
- UCD is a **collaborative** process that **involves** users in the design and development of a system.
- UCD is used to create systems that are **usable**, **efficient**, and **satisfying** to users.



User Story Lifecycle



Outline

- 1 Concepts Generation & Selection
- 2 Basic Concepts
- 3 Requirements Engineering



What is Requirements Engineering?

- **Requirements engineering** is the process of eliciting, analyzing, specifying, validating, and managing the requirements of a system.

- It is a critical activity in the systems development lifecycle that ensures that the system meets the needs of its users.

- It is a collaborative process that involves stakeholders from different backgrounds and perspectives.

understanding

client approval

agile

all project time life

theory



What is Requirements Engineering?

- **Requirements engineering** is the process of eliciting, analyzing, specifying, validating, and managing the requirements of a system.
- It is a critical activity in the systems development lifecycle that ensures that the system meets the needs of its users.
- It is a collaborative process that involves stakeholders from different backgrounds and perspectives.

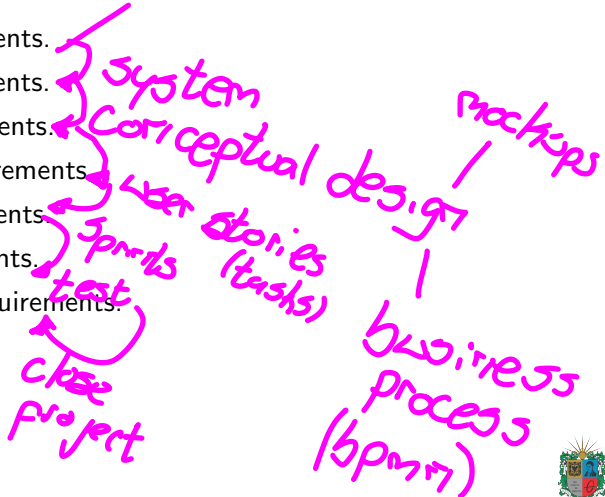
expectations align.



Requirements Engineering Process

The **requirements engineering** process *consists* of the following activities:

- Gathering requirements.
- Analyzing requirements.
- Validating requirements.
- Documenting requirements.
- Managing requirements.
- Verifying requirements.
- Communicating requirements.



Gathering Requirements

- **Gathering** requirements is the process of collecting and documenting the needs of stakeholders.
- It involves interviewing stakeholders, conducting surveys, and observing users to understand their requirements.
- It is essential to prioritize requirements based on stakeholder feedback and project goals.

Problem

Final Product } Product discovery



Clients Are Not Always Right

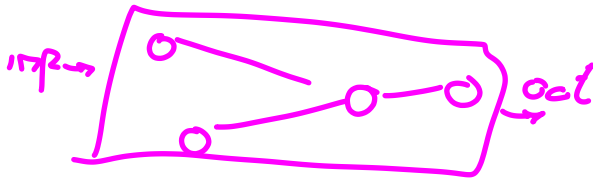
Dear Santa,
 How are you? I'm good.
 Here is what I want for
 Christmas.

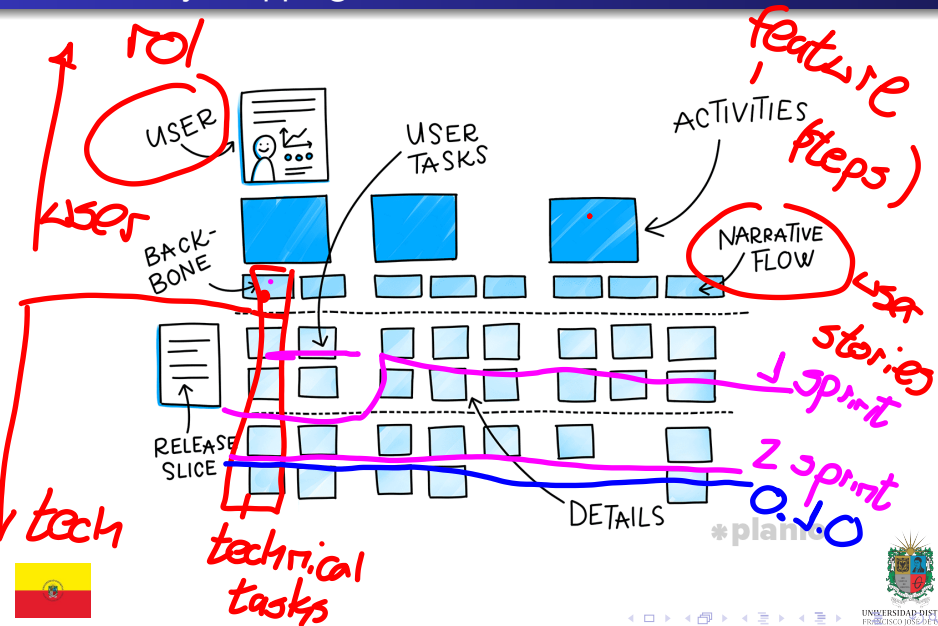
A http://www.amazon.com/gp/product/B0032HFG0M/ref=ssq_hps_bw_g21-ir03?pf_rd_m=ATVPDKIKX0DER&pf_rd_s=center-3&pf_rd_e=gxwy42FH2K03Y7BMWQNM&pf_rd_t=101&pf_rd_p=1328901542&pf_rd_i=165379



Analyzing Requirements

- **Analyzing** requirements is the process of examining and understanding the requirements of a system. → *diagram*
- It involves identifying dependencies, conflicts, and inconsistencies in the requirements. → *trade-off*
- It is a critical activity that ensures that the requirements are complete, consistent, and correct.





USM: Study Case

User-Story Map: Mobile App Feature for Depositing Checks

NNGROUP.COM NN/g

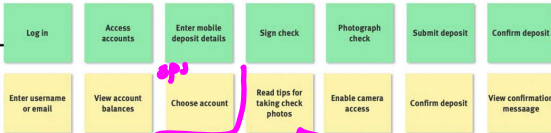
1. Activities:

High-level tasks users can do in the digital product



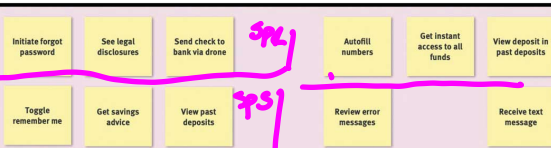
2. Steps:

Steps users go through to complete the activity above



3. Details:

Granular, discrete interactions to complete the step above



Documenting Requirements

- **Documenting** requirements is the process of writing and organizing the requirements of a system.
- It involves creating documents, diagrams, and models that describe the requirements in a clear and concise way.
- It is a collaborative process that involves stakeholders from different backgrounds and perspectives.

→ User stories

→ Mockups

→ Processes
Business

USM

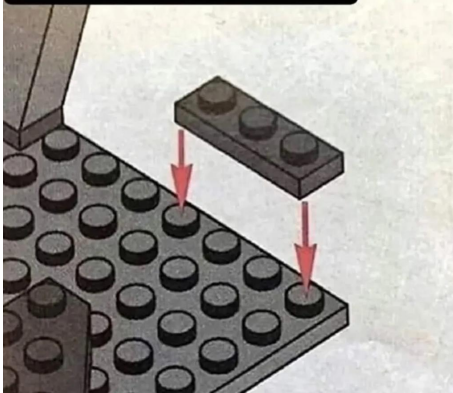


Everyone Hates Writing Documentation

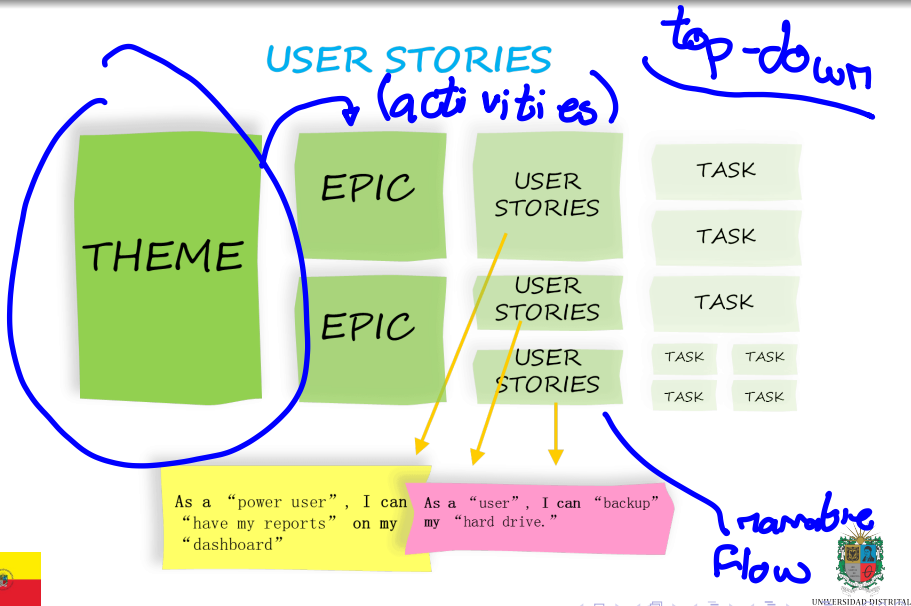
no time
money

**En la documentación
está todo bien explicado**

La documentación:



User Stories Hierarchy



Validating Requirements

- **Validating** requirements is the process of ensuring that the requirements are correct and complete. *⇒ ☺*
- It involves reviewing the requirements with stakeholders to verify that they meet their needs. *Problem*
- It is important to document any changes made during the validation process. *Keep on the radar*
- It is also crucial to review the validation results with stakeholders to ensure alignment with their expectations.



Clients Are Not Always Right



Dad Jokes

@Dadsaysjokes

**

My dad told me his password is:
MickeyMinnieGoofyDonaldPlutoHuey
LouieDeweyDublin.

Because he was told his password
had to contain 8 characters and at
least one Capital.



UNIVERSIDAD DISTRITAL
FRANCISCO JOSÉ DE CALDAS

Verifying Requirements

- **Verifying** requirements is the **process** of **ensuring** that the **requirements** are **correctly implemented** in the **system**.
- It involves **testing** the **system** to **verify** that it **meets** the **requirements**.
- It is a **critical activity** that **ensures** that the **system** is **functional**, **reliable**, and **usable**.



Typical Mistakes When Testing

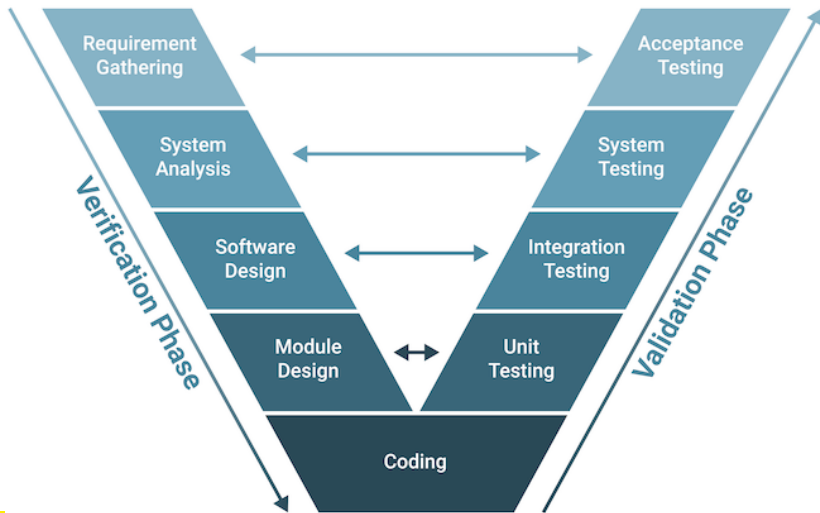
Disturbing Chinese calorie app...



	カシューナッツ (cashew)	1粒	9 kcal
	ジャムパン (Pomeranian)	1個	327 kcal



V-Model in SDLC



Outline

- 1 Concepts Generation & Selection
- 2 Basic Concepts
- 3 Requirements Engineering



Thanks!

Questions?



Repo: www.github.com/EngAndres/ud-public/tree/main/courses/software_engineering_seminar

